



Nebraska Public Power District

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NLS2020035

June 30, 2020

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2020-001-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2020-001-00.

There are no new commitments contained in this letter.

Sincerely,

John Dent, Jr.
Vice President Nuclear -
Chief Nuclear Officer

/jo

Attachment: Licensee Event Report 2020-001-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via ICES entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

IE22
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form <http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Cooper Nuclear Station	2. Docket Number 05000298	3. Page 1 of 4
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4. Title
Reactor Building Doors Opened Simultaneously Could Have Resulted in Loss of Safety Function

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
05	01	2020	2020	- 001	- 00	06	30	2020	Facility Name	Docket Number
										05000
										05000

9. Operating Mode 1	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
10. Power Level 100	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
☐ 50.73(a)(2)(i)(C) ☐ Other (Specify in Abstract below or in NRC Form 366A)				

12. Licensee Contact for this LER	
Licensee Contact Linda Dewhirst, Regulatory Affairs and Compliance Manager	Telephone Number (Include Area Code) (402) 825-5416

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES
A	NG	IEL	P370	Y					

14. Supplemental Report Expected		15. Expected Submission Date		Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date)	☒ No					

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On May 1, 2020, at approximately 08:25, an individual traversed into the Reactor Building through both the inner and outer airlock doors, however, upon checking the inner airlock door to ensure it had latched closed, the door opened. The individual let the door close and checked the door again, with the same result. Upon checking the door a third time, another individual was proceeding to enter the Reactor Building and opened the outer airlock door. At 08:31, an alarm in the control room indicated that both Reactor Building doors were open, resulting in a Secondary Containment breach. Secondary Containment was declared inoperable and Technical Specification (TS) Limiting Condition for Operation (LCO) 3.6.4.1, Condition A was entered. At 08:36, one access door was verified closed, Secondary Containment was declared operable and TS LCO 3.6.4.1, Condition A was exited.

It was discovered that the inner Reactor Building door solenoid locking mechanism was in a degraded condition. At times, the locking solenoid pin was sticking or moving very slowly, but sometimes it worked correctly. The degraded solenoid locking mechanism was replaced and verification of proper operation was completed on May 5, 2020.

This event is considered to have low safety significance. Event Report 54690 was made to the Nuclear Regulatory Commission.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Cooper Nuclear Station	05000298	YEAR	SEQUENTIAL NUMBER	REV NO.
		2020	- 001	- 00

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station was in Mode 1, Power Operation, at 100 percent power at the time of the event on May 1, 2020.

BACKGROUND

Secondary Containment serves as a barrier to confine and monitor potential releases during fuel handling operations and is a system that limits the release of radioactive materials to the environment and consists of four subsystems. One of the subsystems is the Reactor Building [EIS: NG], which encloses the Reactor Pressure Vessel [EIS: RPV] and Primary Containment [EIS: NH].

The Reactor Building is designed to provide protection for the engineered safeguards and nuclear safety systems located in the building from postulated environmental events.

Sets of sealed doors [EIS: DR] (inner and outer doors) are provided for personnel and equipment access. One set of doors in each penetration [EIS: PEN] is required to be closed to maintain Secondary Containment. In addition, the personnel access airlock doors are equipped with interlocks [EIS: IEL] so that one door cannot be opened unless the second door is closed.

The interlocked door can be "by-passed" by actuating the Interlock Bypass Switch or by using the panic bar to open the adjacent inactive leaf.

Reactor Building personnel airlock doors R101 and R102 form part of the Secondary Containment boundary.

EVENT DESCRIPTION

On May 1, 2020, at approximately 08:25, an individual traversed into the Reactor Building through both the inner and outer airlock doors (R101 and R102, respectively); however, when the individual checked the inner airlock door to ensure it had latched closed, the door opened. The individual let the door close without assistance and checked the door a second time, with the same result. Upon checking the door a third time, another individual was proceeding to enter the Reactor Building and opened the outer airlock door, resulting in both doors being opened simultaneously for less than 1 second. At 08:31, an alarm in the control room indicated that both Reactor Building doors were open, resulting in a Secondary Containment breach. As such, Secondary Containment was declared inoperable and Technical Specification (TS) Limiting Condition for Operation (LCO) 3.6.4.1, Condition A was entered. One access door was verified closed making Secondary Containment operable and TS LCO 3.6.4.1, Condition A was exited at 08:36.

Upon investigation, it was discovered that the inner Reactor Building door solenoid locking mechanism was found to be in a degraded condition. The locking solenoid pin was found to be sticking or very slowly moving, but sometimes it worked correctly.

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Communication was shared with the site regarding the condition of the Reactor Building doors and entry into the Reactor Building. This communication included information on how to traverse through the doors to ensure risk of both doors being open was mitigated, and to only enter the Reactor Building for scheduled work until further notice.

The degraded solenoid locking mechanism was replaced and verification of proper operation was completed on May 5, 2020.

BASIS FOR REPORT

This event is reportable as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

An engineering review was performed, and the results demonstrate Secondary Containment remained capable of fulfilling its safety function for the duration of the documented event. In the event of an accident, the Standby Gas Treatment (SGT) system would have automatically started and would have been able to maintain vacuum on Secondary Containment as assumed in the Loss of Coolant Accident dose analysis. As such, this event will not be counted as a Safety System Functional Failure for the Nuclear Regulatory Commission performance indicator since no loss of safety function occurred.

Event Notification 54690 was made to the Nuclear Regulatory Commission.

SAFETY SIGNIFICANCE

This event is considered to have low safety significance. Having both Reactor Building doors open simultaneously does not have a direct or indirect impact on the frequency of core damage. The open flow path into the Reactor Building during SGT system operation will affect the ability of the SGT system to reduce Reactor Building pressure below atmospheric pressure. Due to the duration of the event being less than one second, this is not a concern because the limiting case evaluation in the engineering analysis assumes the SGT system starts 16 seconds after the event initiates.

CAUSE

Opening of the outer door when the inner door was already open due to checking the inner door after traversing through to the Reactor Building was the direct cause. The solenoid locking mechanism in door R101 failed to operate as designed. Specifically, the locking pin did not move to the extended position in a timely manner, resulting in the inner door not providing a positive barrier when the outage door was opened. A causal factor was identified that there is no preventive maintenance (PM) strategy to identify or eliminate degraded conditions for the solenoid locking mechanisms of the doors, resulting in maintenance being performed only when a failure is recognized. An additional causal factor identified that previous degraded performance of the inner door was not recognized or communicated to mitigate the risk of the solenoid locking mechanism failure.

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CORRECTIVE ACTIONS

With the door locking mechanism in a degraded condition, communication was made to the site which included information on how to traverse through the doors to ensure risk of both doors being opened was mitigated and to only enter the Reactor Building for scheduled work until further notice. This communication was included in pre-job briefs involving work being performed in the Reactor Building to ensure personnel understood the risks associated with traversing into and out of the Reactor Building.

The R101 solenoid locking mechanism was replaced. A PM strategy for solenoid locking mechanisms of doors R101 and R102 will be developed. Communication will be shared with the site about recognizing degrading trends and challenging the risk associated with accepting multiple failures and how to communicate these failures through the process to ensure visibility of risk is realized.

PREVIOUS EVENTS

On August 8, 2019, Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2019-002-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On January 6, 2014, Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2014-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On September 10, 2012, Secondary Containment was breached due to both airlock doors in the Reactor Building being inadvertently left open simultaneously. This was reported under LER 2012-003-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.